

jcm-strat: Phases 0 to 3

Approach A, a stripped-down stratosphere model on JCM / Dinosaur semi-Lagrangian. Status record as of 3 September 2026. Repository [reflective-org/jcm-strat](https://reflective-org.github.io/jcm-strat), working copy [/data/JCM_stripped/jcm-strat](#).

The idea in one paragraph

Start from the full JCM climate model, remove every piece of tropospheric physics, replace what the troposphere needs with ERA5 nudging, then add passive tracers to measure stratospheric transport. Each phase adds or removes exactly one thing, so any change in behaviour has one cause. The target for Part 1 is a model that runs years of stratospheric tracer transport per hour on one H100, conserves tracer mass, stays non-negative, and produces an age-of-air pattern comparable to CLaMS. All runs use GPU 0 of the shared 8x H100 node, one at a time, on the T63L95 grid (1.9 degrees, 95 hybrid levels, top near 0.01 hPa) with a 12 minute time step.

Phase overview

Phase	What changed	Run length	Speed (sim. days / hour, stepping)	Status
0	Nothing. Stock JCM, full ECHAM physics	10 days	52	merged (PR 16)
1	Removed all 12 ECHAM terms; kept Held-Suarez only (dry)	1 year	5 787	PR 21 open
2	Added ERA5 nudging of u, v, T below 150 hPa; ERA5 initial state	1 year (2005)	5 455	PR 22 open
3	Added four passive tracers (aoa, unity, sai, e90)	1 year (2005)	4 458	PR 24 open
4	Nothing physical; run 2005-2009 as five chained years	5 years	~4 470	analysis in progress

Phase 0: reproduce the stock model

Nothing of ours. Stock JCM with the full ECHAM physics package (RRTMGP radiation, Tiedtke convection, clouds, TKE diffusion, surface scheme, Hines and orographic gravity-wave drag) on the T63L95 grid for 10 days. Purpose: prove the environment is sound and set the reference speed.

- Environment: uv-managed Python 3.11, JAX 0.10.2 with CUDA 12, dinosaur 1.4.0 from the pinned semi-lagrangian branch (bd99e39b), JCM 2.1.0b0 from the pinned dev branch (849893b). Everything, including caches and runs, lives under /data because the root disk is 97 percent full.
- Result: 51.6 simulated days per hour at a 12 minute step and 51.9 at 15 minutes. The published A100 reference for this configuration is 52. Wall time per simulated day is independent of the time step because radiation, firing every 2 hours, dominates the cost. The speed lever is therefore removing physics, not lengthening the step.
- One real bug found and fixed: JAX silently ran on CPU until the containerised libcuda path was added to the loader path. The environment check now asserts a CUDA device.

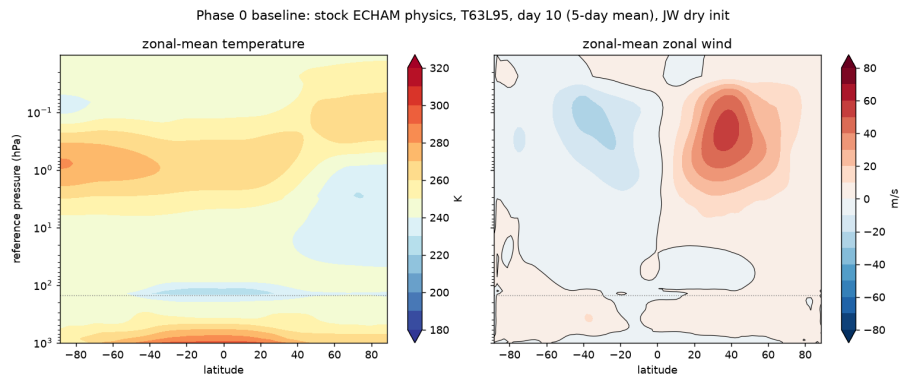


Figure 1. Phase 0, stock JCM with full physics after 10 days: zonal-mean temperature (left) and zonal wind (right). A spin-up from the dry initial state, shown as a sanity check only: no NaNs, plausible vertical structure, jets still forming.

Phase 1: strip the physics (dry Held-Suarez)

Removed: all 12 ECHAM terms. Convection, cloud fraction, cloud microphysics, boundary-layer diffusion, surface exchange, RRTMGP radiation, aerosol, chemistry, and both gravity-wave schemes. Kept: Held-Suarez (1994) only, that is Newtonian relaxation of temperature toward a prescribed profile plus Rayleigh friction near the surface. The model is dry: it carries no water, has no moisture physics, and starts from a zero-humidity Jablonowski-Williamson state.

Held-Suarez acts at every level. Below about the tropopause the equilibrium temperature has real structure (warm tropics, cold poles). Above it the target is clamped to a flat 200 K at every latitude and the relaxation time is 40 days. The stratosphere is therefore pulled toward an isothermal state with no seasonal cycle and no polar-night jet. A separate upper sponge relaxes the top ten levels toward 250 K as a numerical lid.

- One year, no NaNs in any of 13 chunks, surface pressure drift +0.005 hPa, top-level temperature flat.
- Speed: 5 787 simulated days per hour stepping, about 110 times Phase 0. A simulated year takes 9 minutes end to end including compile and output.
- Stratosphere is 21 K RMS colder than ERA5 between 7 and 70 hPa and has essentially no zonal wind. This is the known limitation of Held-Suarez, not a bug; fixes are filed as issues 2 (RRTMGP) and 5 (Polvani-Kushner).
- The planned gravity-wave-drag variant could not be composed with Held-Suarez on JCM's 3-D physics path and was dropped (issue 20). Phase 3 later removed that blocker.

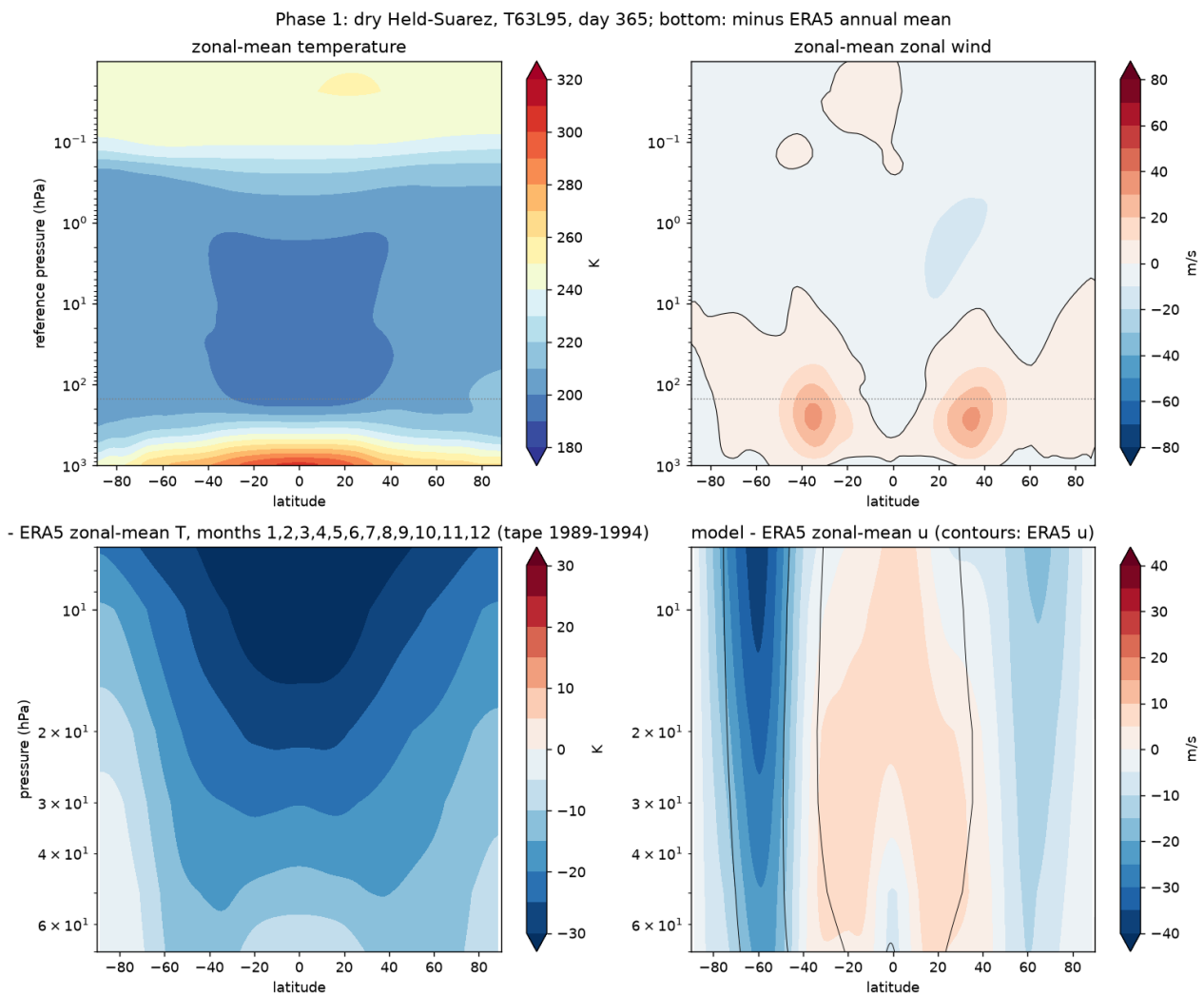


Figure 2. Phase 1 after one year. Top: zonal-mean temperature and zonal wind. Two symmetric subtropical jets near 200 hPa, an isothermal 200 K stratosphere, no polar-night jet. Bottom: model minus the ERA5 1989-1994 annual-mean zonal-mean tape over 7-70 hPa, the stratospheric cold bias and the missing stratospheric winds.

Phase 2: nudge the troposphere to ERA5 (specified dynamics)

Added: the initial state from ERA5 on 1 January 2005, and Newtonian nudging of the zonal wind, meridional wind and temperature toward 6-hourly ERA5 with a 6 hour relaxation time. Nothing else changed. The nudging is masked off in the lowest two model levels (boundary layer free) and at every level with reference pressure below 150 hPa, so the whole stratosphere evolves freely. Surface pressure, humidity and tracers are never nudged.

- One year (2005), stable, surface pressure drift -0.02 hPa, speed within 6 percent of Phase 1.
- The troposphere now carries real 2005 meteorology: global kinetic energy has a seasonal cycle, high in boreal winter and lowest around day 150, which the seasonless Phase 1 could not produce. Jets are asymmetric between hemispheres in late December.
- No visible discontinuity at the 150 hPa cutoff in temperature or wind.
- Stratosphere above 150 hPa is unchanged from Phase 1 by construction: still Held-Suarez, no polar-night jet. Planetary waves from the nudged troposphere can propagate up, but there is no radiatively driven vortex for them to act on.
- Not yet measured: RMSE against ERA5 at 500 hPa, the tropopause position, and the January 2006 SSW. The diagnostic scripts are issue 17.

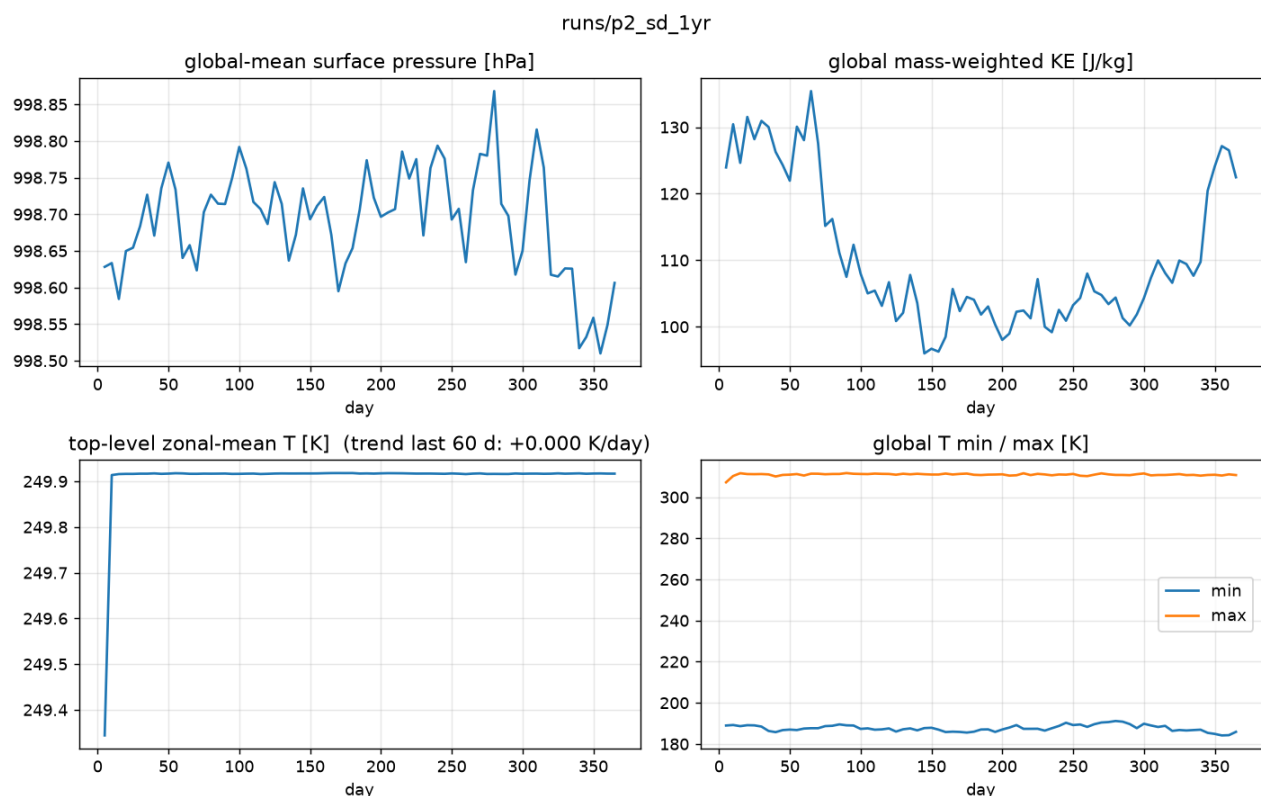


Figure 3. Phase 2 stability summary over 2005: global-mean surface pressure, global kinetic energy, top-level temperature, global temperature extremes. The seasonal cycle in kinetic energy is the signature of the nudged troposphere.

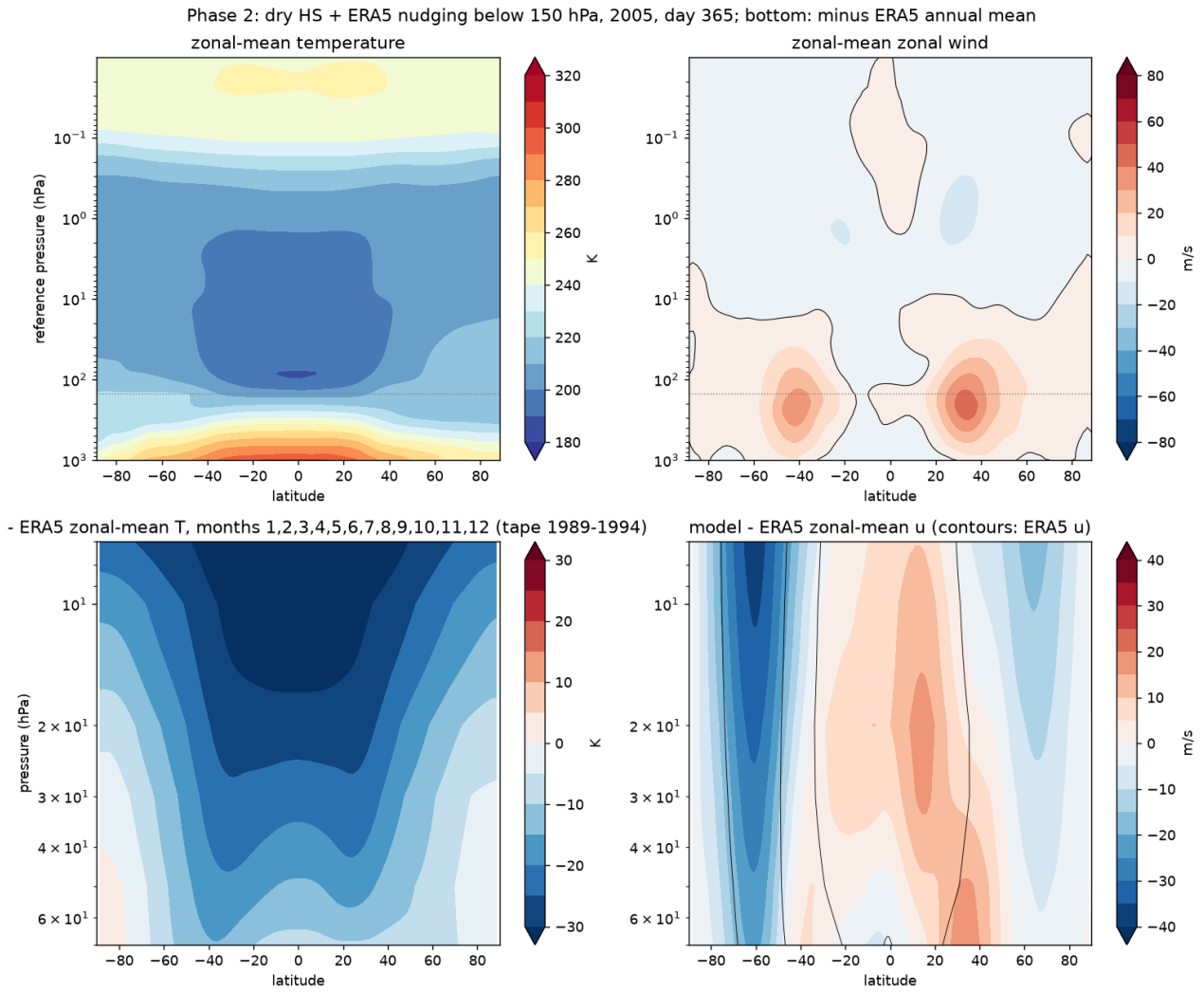


Figure 4. Phase 2 at day 365 (late December 2005). Top: hemispherically asymmetric jets, no kink at 150 hPa (dotted line). Bottom: the stratospheric bias against ERA5 is the same as in Phase 1, as it must be.

Phase 3: passive tracers

Added: one physics term carrying four passive tracers, advected by the semi-Lagrangian dycore as grid-point (nodal) fields with JCM's global proportional mass fixer on. Tracers ride on the winds and never feed back on the dynamics, so the meteorology is identical to Phase 2. Each tracer isolates a different property of the transport scheme:

Tracer	Definition	Question it answers	Result after 1 year
unity	1 everywhere, no sources or sinks	Does the scheme move air without inventing or destroying it? Any deviation is numerical error.	Worst cell off by 2.6e-4. Pass.
sai	Constant source in 15S-15N, 25-55 hPa (about 20-25 km); no sink	Is mass conserved with a known analytic answer (burden = source x box mass x time)? Also: minimum exactly zero, no polar pull-up at the top.	Tracks the analytic line to 1.4 percent; min 0.0; polar top-level value 0.1 percent of the global column. Pass.
aoa	Clock: +1 day per day everywhere, reset to 0 where $p > 700$ hPa	How long has this air been in the stratosphere? The mean age of air, the primary transport metric.	0.97-0.99 yr above 100 hPa: correct but uninformative after one year. Needs Phase 4.
e90	100 in the two lowest layers, 90-day e-folding sink (Prather et al. 2011)	Where is the tropopause? The 90 contour is a standard dynamical tropopause marker.	Does not work here: the dry model has no convection to stir the troposphere, so the 90 contour sits at 970 hPa. Issue 23.

The genuine find. The first attempt lost 56 percent of the clock over the year. JCM's global mass fixer restores each tracer's global mass after every step. At the sharp 700 hPa reset edge the limiter creates a little mass each step, and the fixer removed it by rescaling the whole field, stratosphere included. A clock is not a conserved quantity and must not be mass-fixed. It was invisible in every conservation diagnostic (unity, sai and e90 were all fine). Fixed with a config key that exempts aoa from the fixer, verified by a 5-day run and a unit test.

- Speed: 4 458 simulated days per hour stepping; four extra tracers cost about 1 ms on a 6 ms step. End to end a simulated year takes 10.5 minutes.
- Held-Suarez had to be re-implemented as a per-column variant with identical forcing, because JCM's 3-D physics path cannot add tracer tendencies across terms. This also unblocks gravity-wave drag (issue 20).

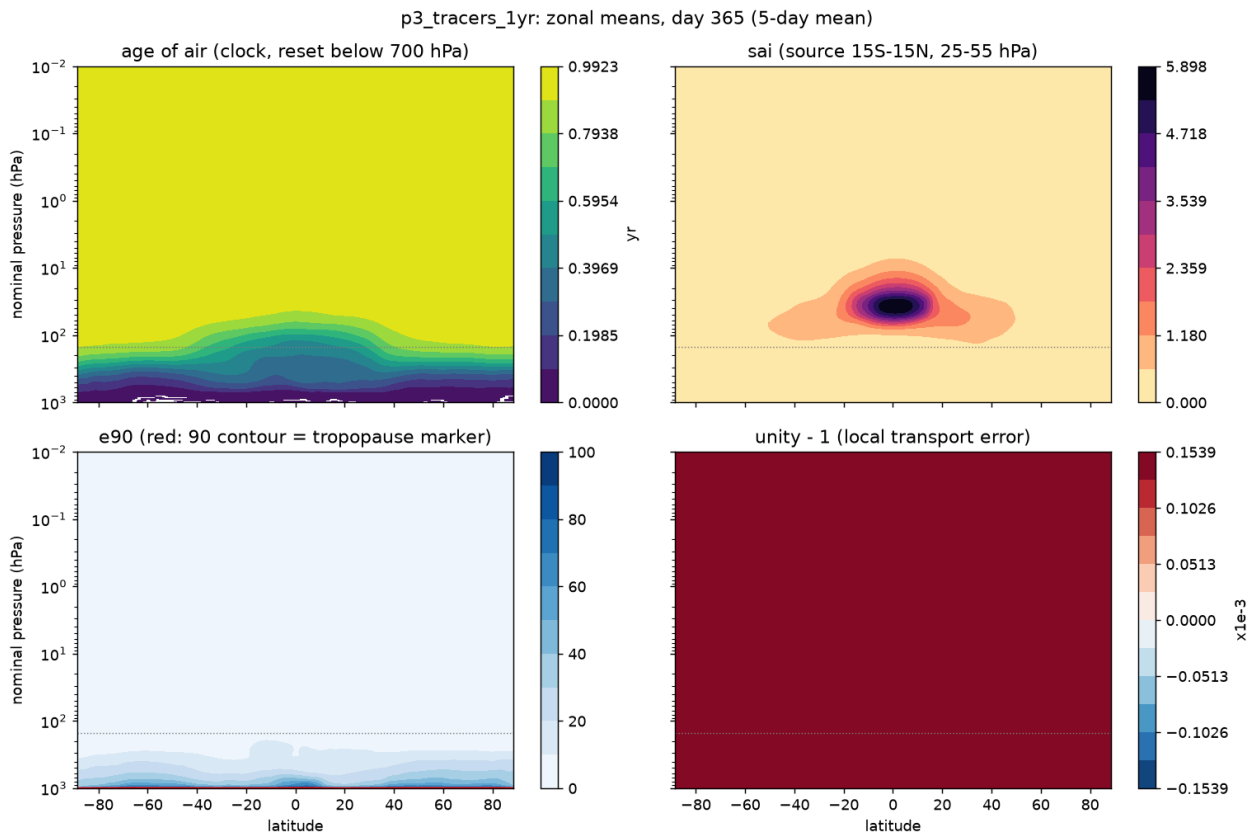


Figure 5. Phase 3 zonal means at day 365. Age of air (top left) reads about one year everywhere above 100 hPa, with only a faint tropical minimum: the stratosphere has aged, fresh air has barely entered. The injection tracer (top right) stays in its source region. e90 (bottom left) decays from the surface instead of marking the tropopause. unity - 1 (bottom right) is uniform to $1.5\text{e-}4$.

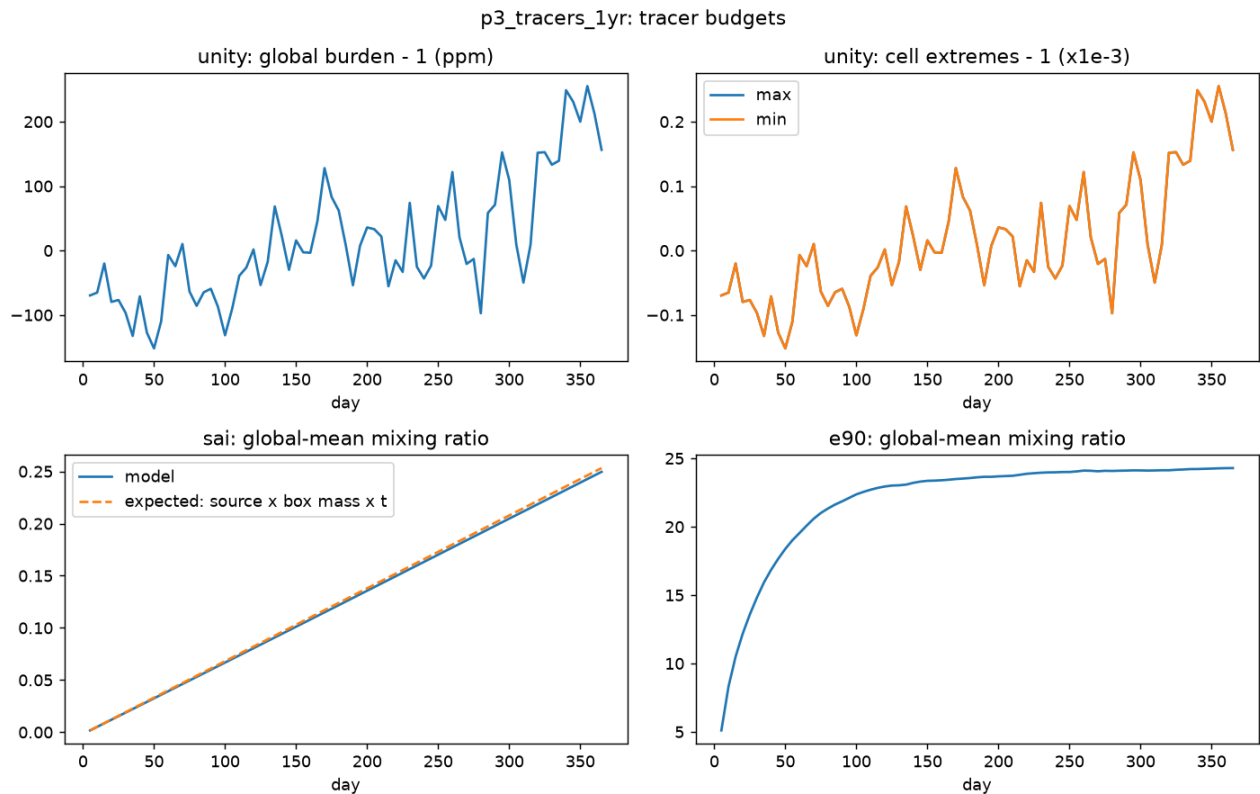


Figure 6. Phase 3 budgets over 2005. unity's global burden and cell extremes stay within $3\text{e-}4$ of one. The sai burden (bottom left) follows the analytic source line. e90's global mean saturates after about 100 days as expected for a 90-day lifetime.

Speed across the phases

Blue is model stepping only, from per-chunk wall time. Orange is end to end including JIT compile and writing output. The end-to-end number is the one that matters for a 30-year run; the gap between the two is output volume and per-chunk overhead (issue 19). The 12 minute step is far below the semi-Lagrangian ceiling, so a time-step sweep (issue 3) is the next speed lever.

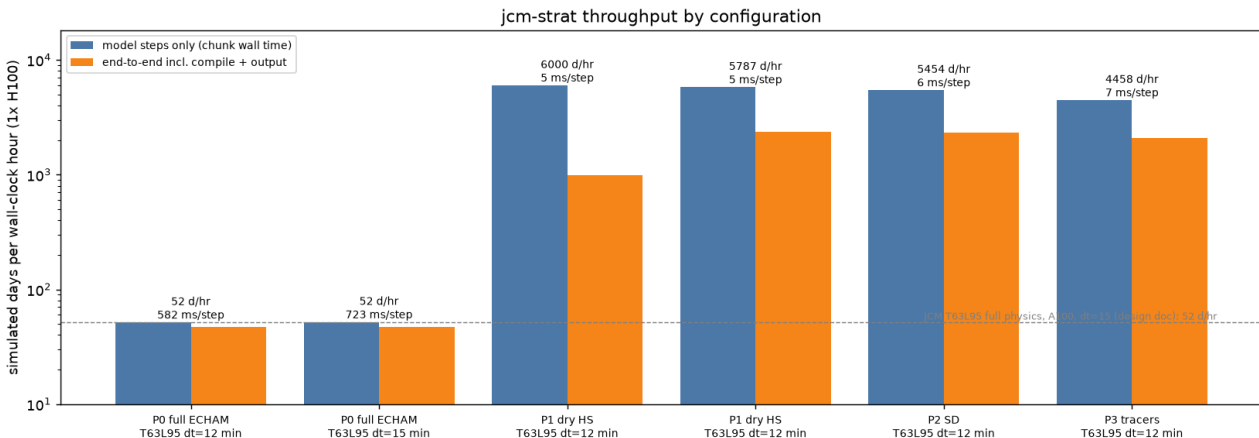


Figure 7. Simulated days per wall-clock hour on one H100 for each configuration, log scale. Removing the physics gave a factor of about 110; nudging cost 6 percent and four tracers a further 18 percent in stepping.

Overall reading

The chain is internally consistent: each phase changed one thing and the diagnostics moved only where they should. The transport machinery is cheap and conserving. The one standing scientific weakness is the Held-Suarez stratosphere, isothermal with no polar-night jet and no seasonal cycle, which will shape the Phase 4 age-of-air pattern and is the first thing to fix afterwards (Polvani-Kushner, issue 5, or RRTMGP with prescribed gases, issue 2).

Reference data: where it comes from

ERA5 for nudging and initialisation (Phases 2 onward)

Source: the WeatherBench2 public ERA5 archive on Google Cloud Storage, anonymous access, read through JCM's own reader (jcm.data.era5). For the T63 grid (192 longitudes) the reader picks the 240 x 121 equiangular store:

`gs://weatherbench2/datasets/era5/1959-2023_01_10-6h-240x121_equiangular_with_poles_conservative.zarr`

- 6-hourly, 1959-2023, 13 pressure levels from 1000 to 50 hPa. Values above 50 hPa are clamped, which is why nudging must stop well below that; the 150 hPa cutoff keeps a wide margin.
- Fields used: u, v, T (nudging target) and the full state for the 1 January 2005 initial condition. Regrided once to the model's hybrid levels and cached under the repository (cache/era5). One year of target is 31 GB and is held on the GPU during the run; five years (154 GB) do not fit an 80 GB card, hence the chained one-year segments in Phase 4.
- Period 2005-2009: volcanically quiet, overlaps the CLaMS record (2004-2023), contains three observed sudden stratospheric warmings (January 2006, February 2008, January 2009).

ERA5 zonal-mean tape for the stratospheric comparison rows

The bottom rows of Figures 2 and 4 compare against a monthly zonal-mean ERA5 tape from the Copernicus Climate Data Store product reanalysis-era5-pressure-levels-monthly-means, prepared earlier for the AIDE atmosphere validation. It holds zonal-mean U and T for 1989-1994 on six levels between 7 and 70 hPa at 0.25 degree latitude. Location:

/home/susanne/docs/AIDE-atmosphere_validation/AIDE-atmosphere/output/era5_monthly_tape.nc

Note that the tape years (1989-1994) differ from the model years (2005). Since the Held-Suarez stratosphere has no interannual variability this is a bias check, not a year-matched comparison.

CLaMS age of air (Phase 4 reference)

Chemical Lagrangian Model of the Stratosphere, version 3.1 (Konopka et al. 2025), driven by ERA5, from the Zenodo dataset 'Gridded CLaMS Simulations Driven by Multiple Reanalyses' (doi:10.5281/zenodo.17357000, PI F. Ploeger, Forschungszentrum Juelich). Monthly zonal means on 39 pressure levels and a 1 degree latitude grid, 2004-2023. The variable AGE is mean age from a clock tracer increasing linearly at the Earth's surface, in years. Local copy:

/data/CLaMS/CLaMS_v3/clams_v3.1_era5_zm_lat.zip (one file per year, e.g. clams_v3.1_era5_zm_lat_press_2005.nc)

Note the definitional difference from our clock: CLaMS resets at the surface, jcm-strat resets everywhere below 700 hPa, and WACCM (below) measures age relative to an entry point at about 100 hPa. The three are compared as patterns, and the analysis script offsets for the entry age.

WACCM6 age of air (Phase 4 reference)

CESM2 WACCM6 REF-D1 simulations (the CCMI-2022 reference historical experiment with specified dynamics), ensemble member 04, AOA1 clock tracer, 1970-2019, 70 levels, 192 latitudes, noleap calendar. Age is given in years relative to a base point at latitude 0, 100 hPa. Members 01-03 and SF6-derived ages are also available in the SAVE subdirectory. Local copy:

/data/CESM2_REFD1_AOA/AoA_waccm6_refd1.04_AOA1mf_1970-2019_ba_0_100.0_ck_0_50.0.nc

Model code and pins

- JCM (climate-analytics-lab/jax-gcm), branch dev, commit 849893b, 1 September 2026, as a submodule.
- Dinosaur (shoyer/dinosaur), branch semi-lagrangian, commit bd99e39b, 24 August 2026, as a submodule; the open upstream PR is [neuralgcm/dinosaur#135](#).
- Every phase's exact command, resolved configuration, commit and wall-clock time is in docs/outputs//output.md in the repository, with the plots shown here beside it.